Appendix B — Recommendation Engine Examples

This appendix provides practical examples of how the recommendation engine should behave based on customer spending patterns.

These examples can be used in:

- Sprint 2 recommendation engine implementation
- Sprint 2 threshold testing
- Sprint 3 dashboard integration
- Sprint 4 showcase demonstrations

The recommendation engine converts spending behaviour into:

- carbon reduction advice
- cost-saving advice
- behaviour improvement suggestions

The MVP recommendation engine is **rule-based**.

AI-driven recommendations are a stretch objective.

Recommendation Logic Model

Basic rule model:

IF SpendAmount > Threshold
THEN Generate Recommendation

Savings model:

$$ProjectedSavings = SpendAmount \times ReductionPercentage$$

Carbon model:

$$CarbonImpact = SpendAmount \times EmissionFactor$$

Travel Recommendation Examples

Threshold:

£300

Example 1 — High Commuter

Customer spend:

£520

Rule triggered:

Travel spend exceeds threshold.

Recommendation:

- reduce solo car journeys
- car share twice per week
- switch one commute day to rail

Estimated savings:

£80 per month

Estimated carbon reduction:

35kgCO2

Example 2 — Family Household

Customer spend:

£450

Recommendation:

- combine family journeys
- reduce unnecessary short trips
- plan weekly shopping trips

Estimated savings:

£60 per month

Estimated carbon reduction:

22kg CO2

Food Recommendation Examples

Threshold:

£250

Example 1 — Single Professional

Customer spend:

£320

Recommendation:

- reduce takeaway meals
- meal prep three times per week
- batch cook lunches

Estimated savings:

£45 per month

Estimated carbon reduction:

12kg CO2

Example 2 — Family Household

Customer spend:

£650

Recommendation:

- plan weekly meals
- reduce food waste
- bulk-buy essentials

Estimated savings:

£90 per month

Estimated carbon reduction:

25kg CO2

Utilities Recommendation Examples

Threshold:

£150

Example 1 — Retired Couple

Customer spend:

£180

Recommendation:

- reduce heating by 1°C
- improve insulation

- reduce standby appliance usage

Estimated savings:

£30 per month

Estimated carbon reduction:

10kg CO2

Example 2 — Family Household

Customer spend:

£260

Recommendation:

- reduce tumble dryer use
- optimise heating schedule
- switch off high-energy devices

Estimated savings:

£55 per month

Estimated carbon reduction:

18kg CO2

Combined Recommendation Example

Customer:

High Commuter

Spend:

Travel: £520
Food: £340
Utilities: £160

Triggered categories:

- Travel
- Food
- Utilities

Combined recommendations:

Travel:

- car share
- rail commute

Food:

- reduce takeaway meals
- meal prep

Utilities:

- reduce heating
- switch off devices

Total projected savings:

£145 per month

Total projected carbon reduction:

57 kg CO2

Sample C# Recommendation Logic

```
if (category == "Travel" && spend > 300)
{
   return "Reduce car journeys and use public transport";
}

if (category == "Food" && spend > 250)
{
   return "Reduce takeaway meals and meal prep at home";
}

if (category == "Utilities" && spend > 150)
{
   return "Reduce heating usage and optimise energy usage";
}
```

Stretch Recommendation Options

Only attempt after MVP completion.

AI Recommendation Option

Using:

Microsoft Power Apps

Examples:

- trend-based recommendations
- behaviour scoring
- customer carbon profile scoring
- personalised challenge recommendations

Trend-Based Rules

Examples:

If travel spend increases for 3 months:

- recommend hybrid work

If food spend spikes:

- recommend meal planning

If utilities rise in winter:

- recommend heating optimisation

Dashboard Display Example

Display:

- category spend
- category carbon impact
- recommendation text
- projected savings
- projected carbon reduction

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